STAC and Sentinel-2 COGs

SpatioTemporal Asset Catalogs
+
Sentinel-2 Cloud-Optimized GeoTiffs

ESIP Summer Meeting 2020

Matthew Hanson
Element 84
@GeoSkeptic

https://tinyurl.com/esip-stac



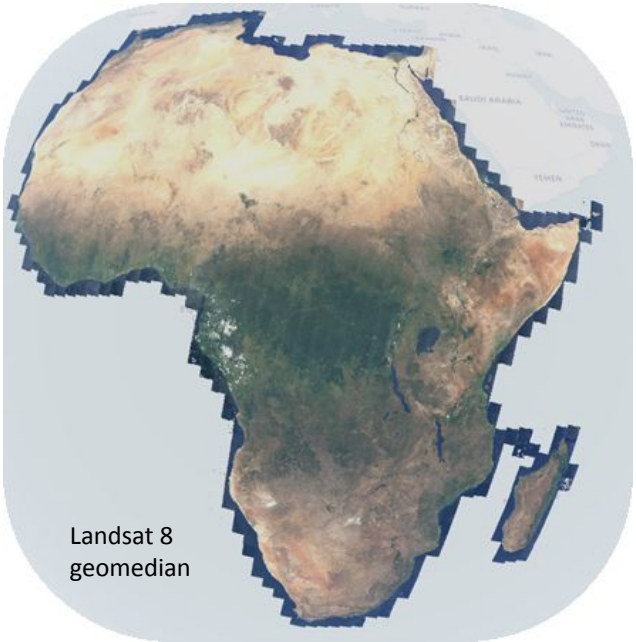
@GeoscienceAus



@DEarthAfrica

Element 84

Digital Earth AFRICA



- 3-year establishment phase funded by
  - Australian Government
  - Helmsley Charitable Trust
- To be hosted in Africa and operated by Africa

For better understanding and addressing challenges of sustainable development in Africa, by providing decision-ready information products.

Support improved economic growth through the innovation of new applications, by enabling businesses and industry to more readily access satellite data.

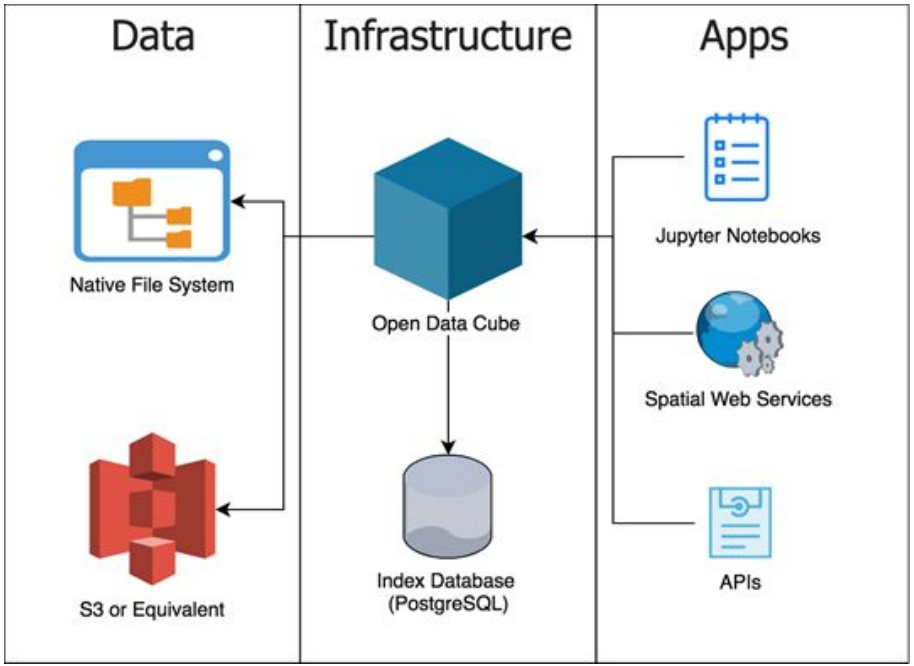
Continental-scale implementations of ODC

Digital Earth AUSTRALIA

NCI AUSTRALIA

National Computational Infrastructure

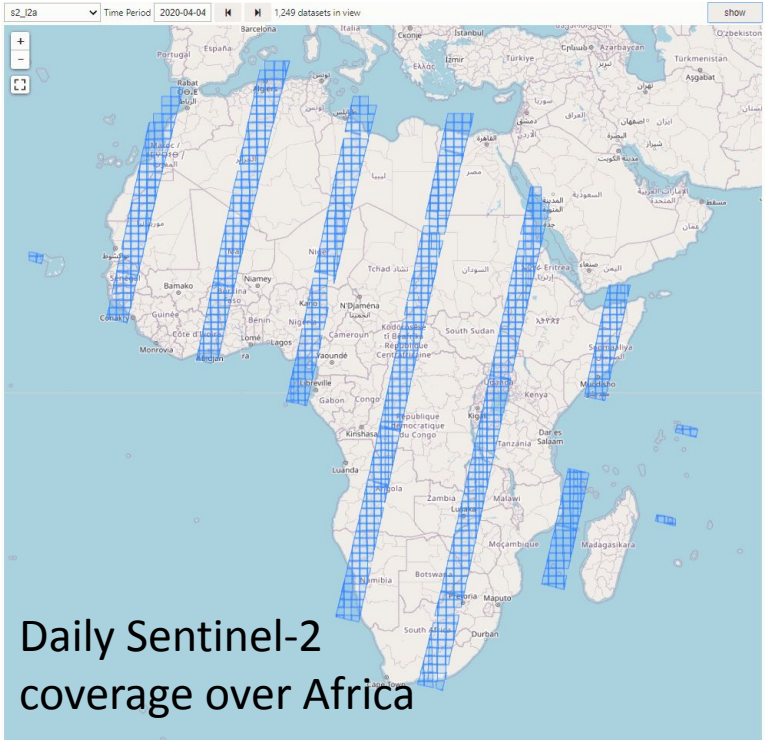
amazon web services



Digital Earth AFRICA

amazon web services

Analysis Ready and Easily Accessible Sentinel-2 for Africa



Daily Sentinel-2 coverage over Africa

COG
CLOUD OPTIMIZED GEOTIFF

STAC
SpatioTemporal Asset Catalog

Being assessed as CEOS Analysis Ready Data

opernicus
Europe's eyes on Earth

SINERGISE

esa

Element 84

amazon web services

Sentinel-2

Sentinel-2A: 06/23/2015 - present

Sentinel-2B: 03/07/2017 - present



10 metre spatial resolution:

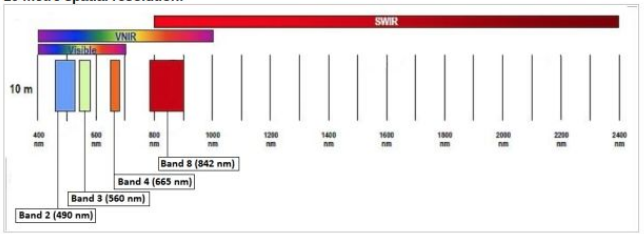


Figure 1: SENTINEL-2 10 m spatial resolution bands: B2 (490 nm), B3 (560 nm), B4 (665 nm) and B8 (842 nm)

20 metre spatial resolution:

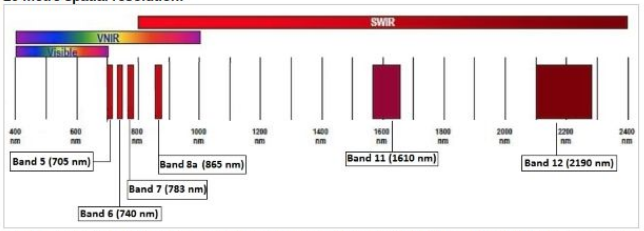


Figure 2: SENTINEL-2 20 m spatial resolution bands: B5 (705 nm), B6 (740 nm), B7 (783 nm), B8a (865 nm), B11 (1610 nm) and B12 (2190 nm)

60 metre spatial resolution:

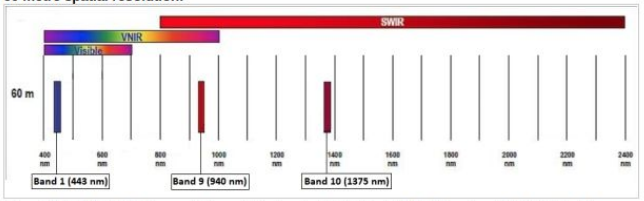
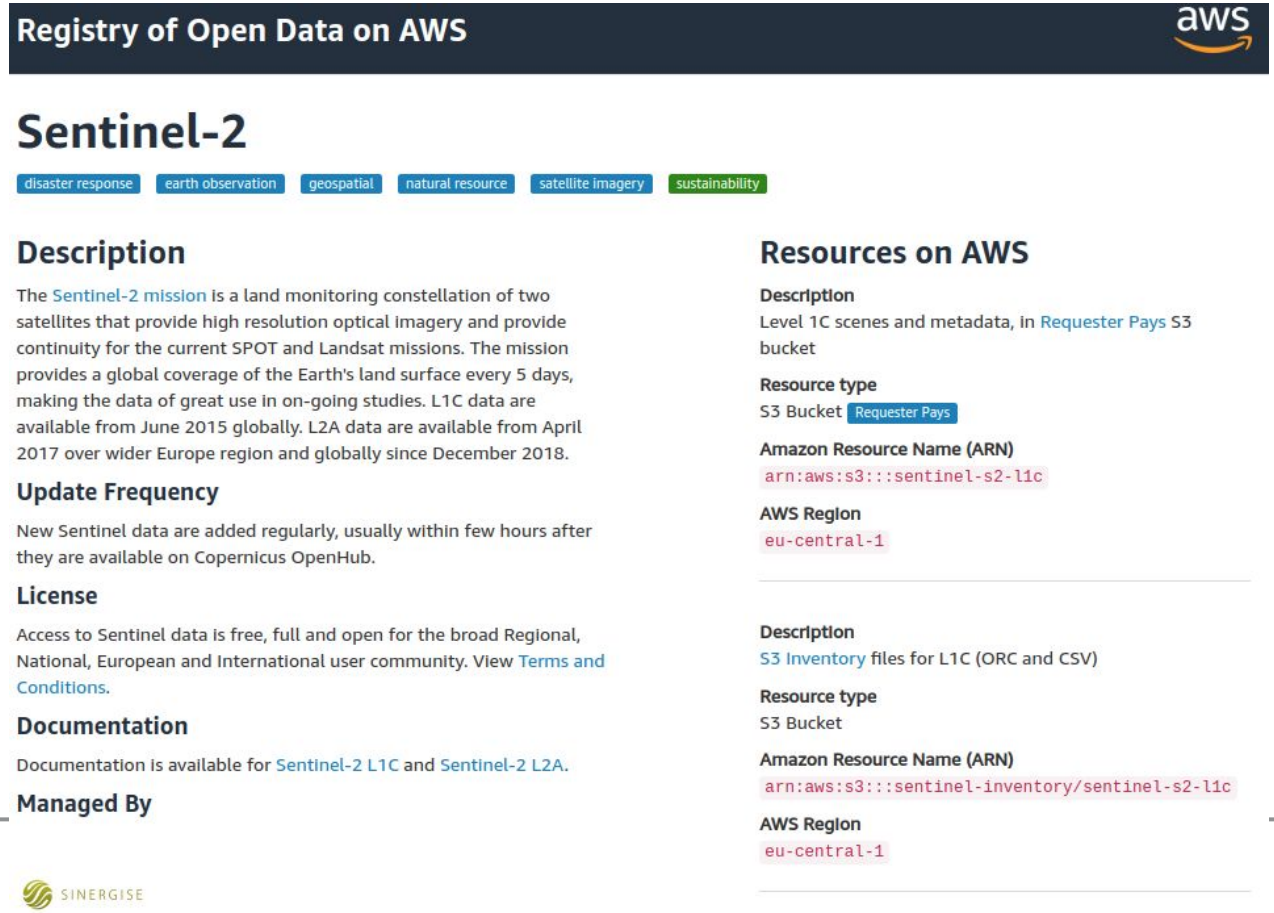


Figure 3: SENTINEL-2 60 m spatial resolution bands: B1 (443 nm), B9 (940 nm) and B10 (1375 nm)

Sentinel-2 Public Datasets



Registry of Open Data on AWS

aws

Sentinel-2

disaster response | earth observation | geospatial | natural resource | satellite imagery | sustainability

Description

The Sentinel-2 mission is a land monitoring constellation of two satellites that provide high resolution optical imagery and provide continuity for the current SPOT and Landsat missions. The mission provides a global coverage of the Earth's land surface every 5 days, making the data of great use in on-going studies. L1C data are available from June 2015 globally. L2A data are available from April 2017 over wider Europe region and globally since December 2018.

Update Frequency

New Sentinel data are added regularly, usually within few hours after they are available on Copernicus OpenHub.

License

Access to Sentinel data is free, full and open for the broad Regional, National, European and International user community. View Terms and Conditions.

Documentation

Documentation is available for Sentinel-2 L1C and Sentinel-2 L2A.

Managed By

Resources on AWS

Description
Level 1C scenes and metadata, in Requester Pays S3 bucket

Resource type
S3 Bucket Requester Pays

Amazon Resource Name (ARN)
`arn:aws:s3:::sentinel-s2-l1c`

AWS Region
`eu-central-1`

Description
S3 Inventory files for L1C (ORC and CSV)

Resource type
S3 Bucket

Amazon Resource Name (ARN)
`arn:aws:s3:::sentinel-inventory/sentinel-s2-l1c`

AWS Region
`eu-central-1`

SINERGISE

Sentinel-2 Products

Description
Level 1C scenes and metadata, in Requester Pays S3 bucket

Resource type
S3 Bucket Requester Pays

Amazon Resource Name (ARN)
`arn:aws:s3:::sentinel-s2-l1c`

AWS Region
`eu-central-1`

Description
S3 Inventory files for L1C (ORC and CSV)

Resource type
S3 Bucket

Amazon Resource Name (ARN)
`arn:aws:s3:::sentinel-inventory/sentinel-s2-l1c`

AWS Region
`eu-central-1`

Description
Level 2A scenes and metadata, in Requester Pays S3 bucket

Resource type
S3 Bucket Requester Pays

Amazon Resource Name (ARN)
`arn:aws:s3:::sentinel-s2-l2a`

AWS Region
`eu-central-1`

Description
S3 Inventory files for L2A (ORC and CSV)

Resource type
S3 Bucket

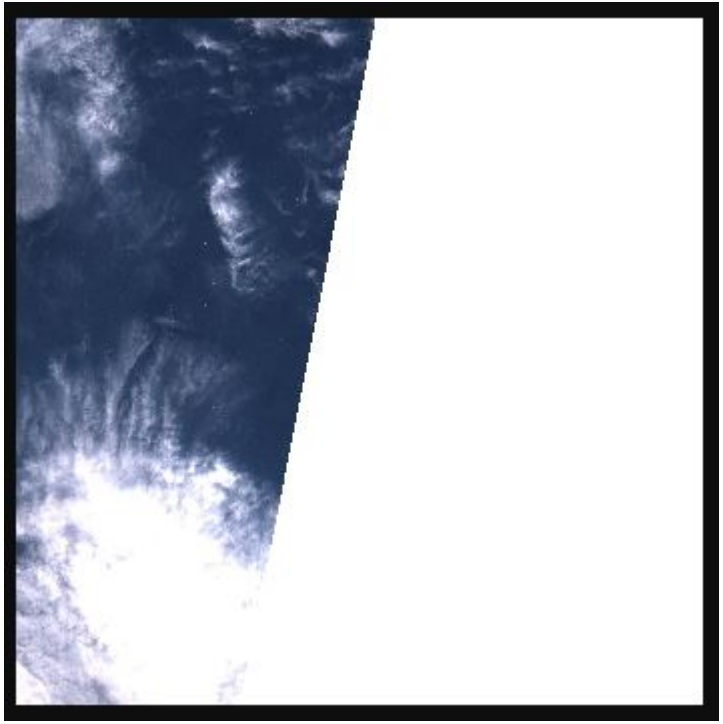
Amazon Resource Name (ARN)
`arn:aws:s3:::sentinel-inventory/sentinel-s2-l2a`

AWS Region
`eu-central-1`

Data Discovery - the old way

```
matthew@aeon:~$ aws s3 ls s3://sentinel-s2-l2a/tiles/59/M/RT/2020/6/11 --request-payer
 --recursive | more
```

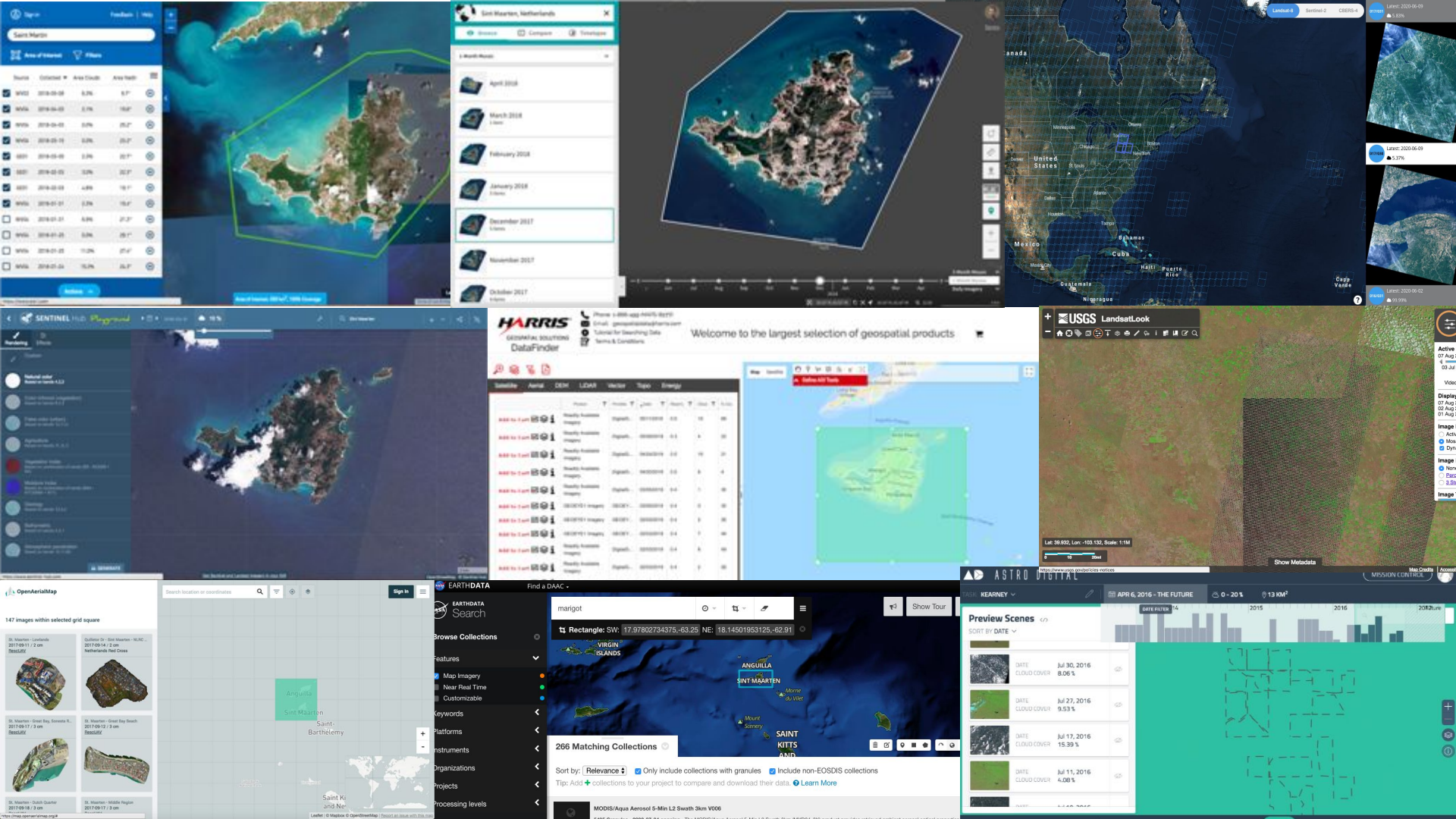
Tile Geometry vs Data Geometry

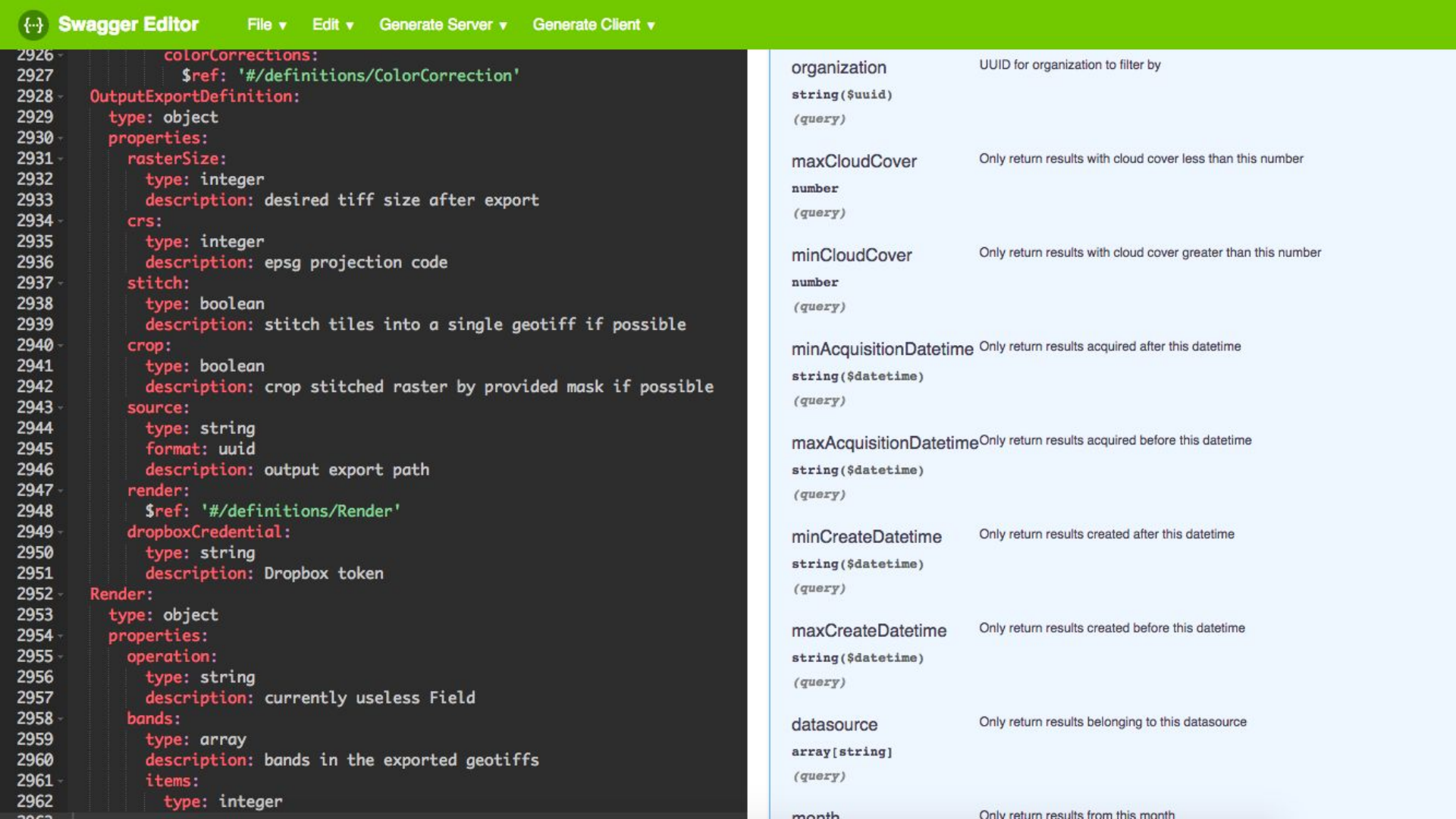


Data Discovery

- Identify specific assets of interest
- Search scenes for valid data within an Area of Interest
- Limit search by date or date range
- Locate only low cloud cover images
- Find scenes from an individual platform (e.g., Sentinel-2B)
- Filter scenes by solar
- Search for scenes higher than a specified resolution

Why STAC?


Saint Martin
Sint Maarten, Netherlands
Welcome to the largest selection of geospatial products
DataFinder
Show Tour
266 Matching Collections
Preview Scenes
Rectangle: SW: 17.97802734375,-63.25 NE: 18.14501953125,-62.91


Swagger Editor
File
Edit
Generate Server
Generate Client
colorCorrections:
$ref: '#/definitions/ColorCorrection'
OutputExportDefinition:
type: object
properties:
rasterSize:
type: integer
description: desired tiff size after export
crs:
type: integer
description: epsg projection code
stitch:
type: boolean
description: stitch tiles into a single geotiff if possible
crop:
type: boolean
description: crop stitched raster by provided mask if possible
source:
type: string
format: uuid
description: output export path
render:
$ref: '#/definitions/Render'
dropboxCredential:
type: string
description: Dropbox token
Render:
type: object
properties:
operation:
type: string
description: currently useless Field
bands:
type: array
description: bands in the exported geotiffs
items:
type: integer
organization
UUID for organization to filter by
string($uuid)
(query)
maxCloudCover
Only return results with cloud cover less than this number
number
(query)
minCloudCover
Only return results with cloud cover greater than this number
number
(query)
minAcquisitionDatetime
Only return results acquired after this datetime
string($datetime)
(query)
maxAcquisitionDatetime
Only return results acquired before this datetime
string($datetime)
(query)
minCreateDatetime
Only return results created after this datetime
string($datetime)
(query)
maxCreateDatetime
Only return results created before this datetime
string($datetime)
(query)
datasource
Only return results belonging to this datasource
array[string]
(query)

Make geospatial specs differently

- Start with a core set of principles to collaborate
  - Embrace open source tools & practices – all work on spec on GitHub
  - JSON/REST/HTTP at the core
  - Small reusable pieces loosely coupled
  - Focus on the developer & implementor – practical & easy to implement
  - “we believe in rough consensus and running code”



What is it?

- SpatioTemporal Asset Catalogs
- Language for geospatial catalogs and assets
- Focus on search and discovery
- Simple + Extensible
- Static catalogs and HTTP APIs

What is not?

- Full-fledged metadata standard
- Single source of truth
- Covering all kind of datasets
- Replacement for ISO 19115, OGC CSW, ...

Who?

- 38 contributors on github
- 16 with 10+ contributions



https://stacspec.org

SpatioTemporal Asset Catalogs

Enabling online search and discovery of geospatial assets

Introduction | Why STAC? | STAC Specification | STAC in Action | STAC Tools | Get Involved

About

The SpatioTemporal Asset Catalog (STAC) specification provides a common language to describe a range of geospatial information, so it can more easily be indexed and discovered. A 'spatiotemporal asset' is any file that represents information about the earth captured in a certain space and time.

The specification

- Catalogs
  - There's always a root catalog
  - Contains other catalogs and collections
- Collections
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Ultimately contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Items
  - A single scene or set of datafiles for a specific location at a specific date and time (when/where)

Links

Field Name	Type	Description
href	string	**REQUIRED.** The actual link in the format of an URL. Relative and absolute links are both allowed.
rel	string	**REQUIRED.** Relationship between the current document and the linked document. See chapter "Relation types" for more information.
type	string	Media type of the referenced entity.
title	string	A human readable title to be used in rendered displays of the link.

Link Relation Types

Type	Description
self	STRONGLY RECOMMENDED. *Absolute* URL to the location that the catalog file can be found online, if available. This is particularly useful when in a download package that includes metadata, so that the downstream user can know where the data has come from.
root	STRONGLY RECOMMENDED. URL to the root STAC Catalog or Collection. Catalogs should include a link to their root, even if it's the root and points to itself.
parent	URL to the parent STAC Catalog or Collection. Non-root catalogs should include a link to their parent.
child	URL to a child STAC Catalog or Collection.
item	URL to a STAC Item.

Other Links: **about, derived-from, previous-version (any IANA rel type)**

STAC Catalog

Element	Type	Description
stac_version	string	**REQUIRED.** The STAC version the catalog implements. STAC versions can be mixed, but please keep the recommended best practices in mind.
stac_extensions	[string]	A list of extension identifiers the Catalog implements.
id	string	**REQUIRED.** Identifier for the catalog.
title	string	A short descriptive one-line title for the catalog.
description	string	**REQUIRED.** Detailed multi-line description to fully explain the catalog. CommonMark 0.29 syntax MAY be used for rich text representation.
links	[Link Object]	**REQUIRED.** A list of references to other documents.

STAC Collection

Element	Type	Description
stac_version	string	**REQUIRED.** The STAC version the Collection implements. STAC versions can be mixed, but please keep the recommended best practices in mind.
stac_extensions	[string]	A list of extension identifiers the Collection implements.
id	string	**REQUIRED.** Identifier for the collection that is unique across the provider.
title	string	A short descriptive one-line title for the collection.
description	string	**REQUIRED.** Detailed multi-line description to fully explain the collection. CommonMark 0.29 syntax MAY be used for rich text representation.
keywords	[string]	List of keywords describing the collection.
license	string	**REQUIRED.** Collection's license(s), either a SPDX License identifier, `various` if multiple licenses apply or `proprietary` for all other cases.
providers	[Provider Object]	A list of providers, which may include all organizations capturing or processing the data or the hosting provider. Providers should be listed in chronological order with the most recent provider being the last element of the list.
extent	Extent Object	**REQUIRED.** Spatial and temporal extents.
summaries	Map<string, [*]|Stats Object>	A map of property summaries, either a set of values or statistics such as a range.
links	[Link Object]	**REQUIRED.** A list of references to other documents.

STAC Item

Field Name	Type	Description
stac_version	string	**REQUIRED.** The STAC version the Item implements.
stac_extensions	[string]	A list of extensions the Item implements.
id	string	**REQUIRED.** Provider identifier. As most geospatial assets are already defined by some identification scheme by the data provider it is recommended to simply use that ID. Data providers are advised to include sufficient information to make their IDs globally unique, including things like unique satellite IDs.
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
geometry	GeoJSON Geometry Object	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Coordinates are specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84.
bbox	[number]	**REQUIRED.** Bounding Box of the asset represented by this item, formatted according to RFC 7946, section 5.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is strongly recommended.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type). This field is *required* if such a relation type is present. This field provides an easy way for a user to search for any Items that belong in a specified Collection.

Item Properties

Field Name	Type	Description
title	string	A human readable title describing the Item.
description	string	Detailed multi-line description to fully explain the Item. CommonMark 0.29 syntax MAY be used for rich text representation.
datetime	string|null	See the Item Spec Fields for more information.
created	string	Creation date and time of the corresponding data (see below).
updated	string	Date and time the corresponding data (see below) was updated last.
start_datetime	string	The first or start date and time for the item, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.
end_datetime	string	The last or end date and time for the item, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.
platform	string	Unique name of the specific platform to which the instrument is attached.
instruments	[string]	Name of instrument or sensor used (e.g., MODIS, ASTER, OLI, Canon F-1).
constellation	string	Name of the constellation to which the platform belongs.
mission	string	Name of the mission for which data is collected.
gsd	number	Ground Sample Distance at the sensor.

Item Assets

Field Name	Type	Description
href	string	**REQUIRED.** Link to the asset object. Relative and absolute links are both allowed.
title	string	The displayed title for clients and users.
description	string	A description of the Asset providing additional details, such as how it was processed or created.
type	string	Media type of the asset.
roles	[string]	The semantic role of the asset, similar to the use of `rel` in links.

Content Extensions

Extension Title	Identifier	Field Name Prefix	Scope	Maturity	Description
Checksum	checksum	checksum	Item, Catalog, Collection	*Proposal*	Provides a way to specify file checksums for assets and links in Items, Catalogs and Collections.
Collection Assets	collection-assets	-	Collection	*Proposal*	Provides a way to specify assets available on the collection-level.
Data Cube	datacube	cube	Item, Collection	*Proposal*	Data Cube related metadata, especially to describe their dimensions.
Electro-Optical	eo	eo	Item	*Proposal*	Covers electro-optical data that represents a snapshot of the earth for a single date and time. It could consist of multiple spectral bands, for example visible bands, infrared bands, red edge bands and panchromatic bands. The extension provides common fields like bands, cloud cover, gsd and more.
Item Asset Definition	item-assets	-	Collection	*Proposal*	Provides a way to specify details about what assets may be found in Items belonging to a collection.
Label	label	label	Item	*Proposal*	Items that relate labeled AOIs with source imagery
Point Cloud	pointcloud	pc	Item	*Proposal*	Provides a way to describe point cloud datasets. The point clouds can come from either active or passive sensors, and data is frequently acquired using tools such as LiDAR or coincidence-matched imagery.

Content Extensions

Projection	projection	proj	Item	*Proposal*	Provides a way to describe items whose assets are in a geospatial projection.
SAR	sar	sar	Item	*Proposal*	Covers synthetic-aperture radar data that represents a snapshot of the earth for a single date and time.
Satellite	sat	sat	Item	*Proposal*	Satellite related metadata for data collected from satellites.
Scientific	scientific	sci	Item, Collection	*Proposal*	Scientific metadata is considered to be data that indicate from which publication data originates and how the data itself should be cited or referenced.
Single File STAC	single-file-stac	-	Catalog	*Proposal*	An extension to provide a set of Collections and Items within a single file catalog.
Tiled Assets	tiled-assets	tiles	Item, Catalog, Collection	*Proposal*	Allows to specify numerous assets using asset templates via tile matrices and dimensions.
Timestamps	timestamps	-	Item	*Proposal*	Allows to specify numerous timestamps for assets and metadata.
Versioning Indicators	version	-	Item, Collection	*Proposal*	Provides fields and link relation types to provide a version and indicate deprecation.
View Geometry	view	view	Item	*Proposal*	View Geometry adds metadata related to angles of sensors and other radiance angles that affect the view of resulting data

STAC Extensions used for Sentinel-2

- EO
- Satellite
- View Geometry
- Projection

Electro-Optical Extension

Properties

Field Name	Type	Description
eo:gsd	number	**REQUIRED.** Ground Sample Distance at the sensor.
eo:bands	[Band Object]	**REQUIRED.** This is a list of the available bands where each item is a Band Object.
eo:cloud_cover	number	Estimate of cloud cover as a percentage (0-100) of the entire scene. If not available the field should not be provided.

Bands object

Field Name	Type	Description
name	string	The name of the band (e.g., "B01", "B02", "B1", "B5", "QA").
common_name	string	The name commonly used to refer to the band to make it easier to search for bands across instruments. See the list of accepted common names.
description	string	Description to fully explain the band. CommonMark 0.29 syntax MAY be used for rich text representation.
center_wavelength	number	The center wavelength of the band, in micrometers (μm).
full_width_half_max	number	Full width at half maximum (FWHM). The width of the band, as measured at half the maximum transmission, in micrometers (μm).

Spectral Band Common Names

Common Name	Band Range (µm)	Landsat 5/7	Landsat 8	Sentinel 2	MODIS
coastal	0.40 - 0.45		1	1	
blue	0.45 - 0.50	1	2	2	3
green	0.50 - 0.60	2	3	3	4
red	0.60 - 0.70	3	4	4	1
yellow	0.58 - 0.62				
pan	0.50 - 0.70	8 (*L7 only*)	8		
rededge	0.70 - 0.75				
nir	0.75 - 1.00	4		8	2
nir08	0.75 - 0.90		5	8a	
nir09	0.85 - 1.05			9	
cirrus	1.35 - 1.40		9	10	26
swir16	1.55 - 1.75	5	6	11	6
swir22	2.10 - 2.30	7	7	12	7
lwir	10.5 - 12.5	6			
lwir11	10.5 - 11.5		10		31
lwir12	11.5 - 12.5		11		32

Satellite and View Geometry

Field Name	Type	Description
sat:orbit_state	string	The state of the orbit. Either `ascending` or `descending` for polar orbiting satellites, or `geostationary` for geosynchronous satellites
sat:relative_orbit	integer	The relative orbit number at the time of acquisition.

Field Name	Type	Description
view:off_nadir	number	The angle from the sensor between nadir (straight down) and the scene center. Measured in degrees (0-90).
view:incidence_angle	number	The incidence angle is the angle between the vertical (normal) to the intercepting surface and the line of sight back to the satellite at the scene center. Measured in degrees (0-90).
view:azimuth	number	Viewing azimuth angle. The angle measured from the sub-satellite point (point on the ground below the platform) between the scene center and true north. Measured clockwise from north in degrees (0-360).
view:sun_azimuth	number	Sun azimuth angle. From the scene center point on the ground, this is the angle between truth north and the sun. Measured clockwise in degrees (0-360).
view:sun_elevation	number	Sun elevation angle. The angle from the tangent of the scene center point to the sun. Measured from the horizon in degrees (0-90).

Projection Extension

Field Name	Type	Description
proj:epsg	integer |null	**REQUIRED.** EPSG code of the datasource
proj:wkt2	string |null	WKT2 string representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:projjson	PROJJSON Object |null	PROJJSON object representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:geometry	Polygon Object	Defines the footprint of this Item.
proj:bbox	[number]	Bounding box of the Item in the asset CRS in 2 or 3 dimensions.
proj:centroid	Centroid Object	Coordinates representing the centroid of the Item in the asset CRS
proj:shape	[integer]	Number of pixels in Y and X directions for the default grid
proj:transform	[number]	The affine transformation coefficients for the default grid

STAC API

STAC API spec
https://github.com/radiantearth/stac-api-spec

OGC API: Features
+
STAC Extensions

OpenAPI Doc: https://stacspec.org/STAC-ext-api.html

OGC API - Features Endpoints

The core OGC API - Features endpoints are shown below, with details provided in an OpenAPI specification document.

Endpoint	Returns	Description
`/`	JSON	Landing page, links to API capabilities
`/conformance`	JSON	Info about standards to which the API conforms
`/collections`	JSON	Object with a list of Collections contained in the catalog and links
`/collections/{collectionId}`	Collection	Returns single Collection JSON
`/collections/{collectionId}/items`	ItemCollection	GeoJSON FeatureCollection-conformant entity of Items in collection
`/collections/{collectionId}/items/{featureId}`	Item	Returns single Item (GeoJSON Feature)

STAC Endpoints

STAC provides some additional endpoints for the root Catalog itself, as well as the capability to search the Catalog. Note that a STAC API does not need to implement OAFeat, in which case it would only support the endpoints given below. See the OpenAPI specification document.

Endpoint	Returns	Description
`/`	Catalog	Extends `/` from OAFeat to return a full STAC catalog.
`/search`	ItemCollection	Retrieves a group of Items matching the provided search predicates, probably containing search metadata from the `search` extension

Search endpoints

Explore STAC

- Install JSONView or other JSON formatter browser extension
- Visit https://earth-search.aws.element84.com/v0
- Explore catalogs
  - sentinel-s2-l1c
  - sentinel-s2-l2a
  - sentinel-s2-l2a-cogs

STAC API Filters

Filter Parameters and Fields

Unless otherwise noted by **Path-only**, these filters are passed as query string parameters, form parameters, or JSON entity fields. For filters that represent a set of values, query and form parameters should use comma-separated string values and JSON entity attributes should use JSON Arrays.

Parameter	Type	APIs	Description
collectionId	string	OAFeat	**Path-only** Single Collection ID to include in the search for items. Only Items in one of the provided Collection will be searched
limit	integer	OAFeat, STAC	The maximum number of results to return (page size). Defaults to 10
bbox	[number]	OAFeat, STAC	Requested bounding box. Represented using either 2D or 3D geometries. The length of the array must be 2*n where n is the number of dimensions. The array contains all axes of the southwesterly most extent followed by all axes of the northeasterly most extent specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84. When using 3D geometries, the elevation of the southwesterly most extent is the minimum elevation in meters and the elevation of the northeasterly most extent is the maximum.
datetime	string	OAFeat, STAC	Single date+time, or a range ('/' seperator), formatted to RFC 3339, section 5.6. Use double dots `..` for open date ranges.
intersects	GeoJSON Geometry	STAC	Searches items by performing intersection between their geometry and provided GeoJSON geometry. All GeoJSON geometry types must be supported.
ids	[string]	STAC	Array of Item ids to return. All other filter parameters that further restrict the number of search results (except `next` and `limit`) are ignored
collections	[string]	STAC	Array of Collection IDs to include in the search for items. Only Items in one of the provided Collections will be searched

STAC API Extensions

Extension Name	Scope*	Description	Maturity
Fields	*None*	Adds parameter to control which fields are returned in the response.	*Pilot*
Query	*None*	Adds parameter to search Item and Collection properties.	*Pilot*
Context	ItemCollection	Adds search related metadata (context) to ItemCollection.	*Proposal*
Sort	*None*	Adds Parameter to control sorting of returns results.	*Pilot*
Transaction	*None*	Adds PUT and DELETE endpoints for the creation, editing, and deleting of items and Collections.	*Pilot*
Items and Collections API Version	*None*	Adds GET versions resource to collections and items endpoints and provides semantics for a versioning scheme for collections and items.	*Proposal*

STAC API Extension

Extension Name	Scope*	Description	Maturity
Fields	*None*	Adds parameter to control which fields are returned in the response.	*Pilot*
Query	*None*	Adds parameter to search Item and Collection properties.	*Pilot*
Context	ItemCollection	Adds search related metadata (context) to ItemCollection.	*Proposal*
Sort	*None*	Adds Parameter to control sorting of returns results.	*Pilot*
Transaction	*None*	Adds PUT and DELETE endpoints for the creation, editing, and deleting of items and Collections.	*Pilot*
Items and Collections API Version	*None*	Adds GET versions resource to collections and items endpoints and provides semantics for a versioning scheme for collections and items.	*Proposal*

```
{
  "fields": {
    "exclude": [
      "geometry"
    ]
  }
}
```

Alternate shortcut

fields=-geometry
fields=properties.eo:cloud_cover

```
{
    "sortby": [
        {
            "field": "properties.created",
            "direction": "asc"
        },
        {
            "field": "properties.eo:cloud_cover",
            "direction": "desc"
        }
```

Alternate shortcut

sortby=-properties.datetime
sortby=+collection

```
{
  "query": {
    "eo:cloud_cover": {
      "gte": 0,
      "lte": 10
    },
    "stringAttr1": {
      "startsWith": "abc",
      "endsWith": "xyz"
```

https://earth-search.aws.element84.com/v0



Earth Search

A Public Search and Discovery API for Earth on AWS Datasets

Cloud-friendly Data Formats

"Hey, let's not download the whole file everytime" - Eugene Cheipesh

- Cloud-optimized GeoTIFF (COG)
- Meta Raster Format (MRF)
- Zarr
- Entwine Point Tiles
- TileDB

Sentinel-2 COGs

- Public dataset of Sentinel-2 L2A
  - Converted to Cloud-Optimized GeoTIFFs
  - All of Africa ~ 1.4 million scenes
  - Global archive coming ~ 9.4 million scenes



Australian Government

Geoscience Australia

Element 84



Digital Earth AFRICA

aws

Sentinel-2 COGs

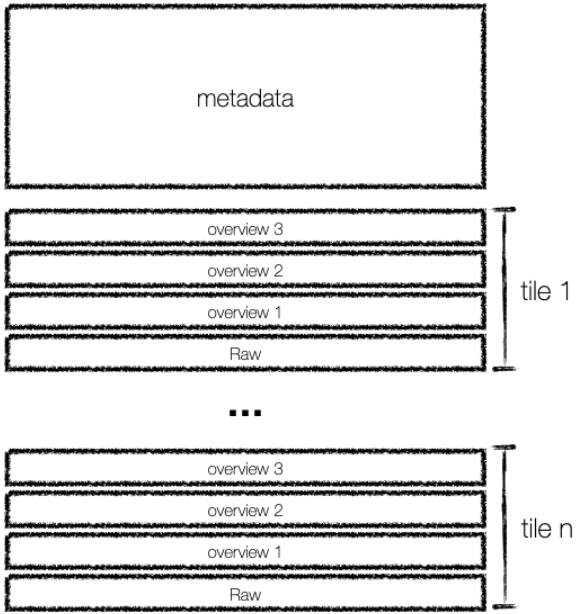
- Sentinel-2 L1C and L2A products in JPEG2000 format
- JPEG2000 is not a `cloud friendly` format

pds	CBERS	LANDSAT	S2 (JP2OpenJPEG)	S2 (JP2KAK)	S2 (JP2ECW)
HTTP call (Z12)	5	5	118	64	115
Bytes transfered (Z12)	53 035	1 722 986	3 920 804	1 310 720	2 741 156
HTTP call (Z10)	6	5	128	74	121
Bytes transfered (Z10)	57 992	934 156	14 390 180	1 196 032	2 053 028
HTTP call (tile)	3	3	113	13	105
Bytes transfered (tile)	49 236	16 915	1 889 188	16 384	1 725 348

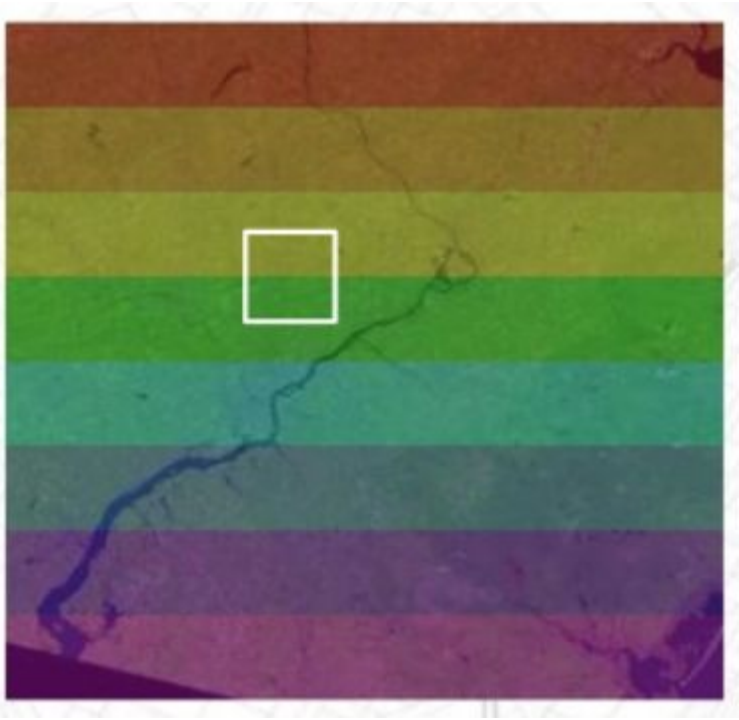
Ref: https://medium.com/@_VincentS_/do-you-really-want-people-using-your-data-ec94cd94dc3f

Cloud-Optimized GeoTIFF

- IFDs (Image File Directory) at beginning of file
  - reduces data seek times
- Uses Tiled Segments
  - decompress and load separately
- Overviews in file
  - efficiently use the data at any resolution

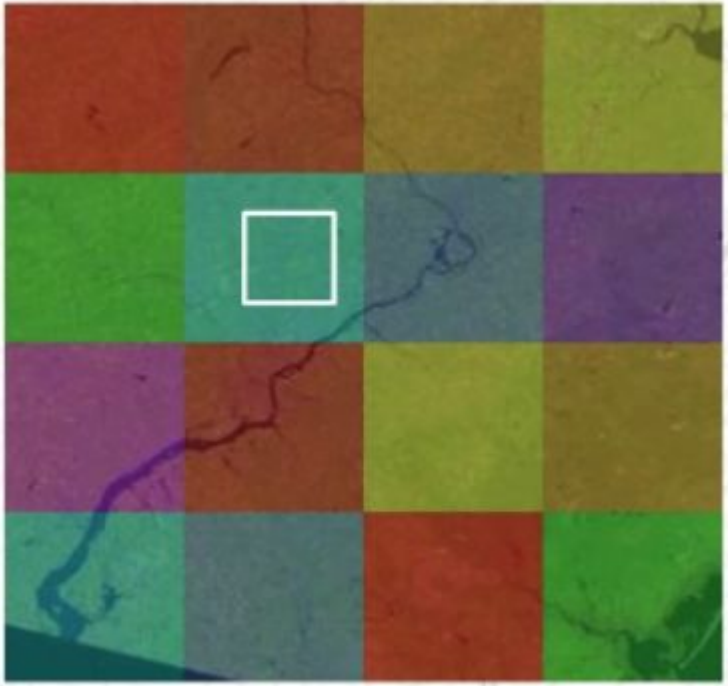


TIFF Striped Segments = BAD



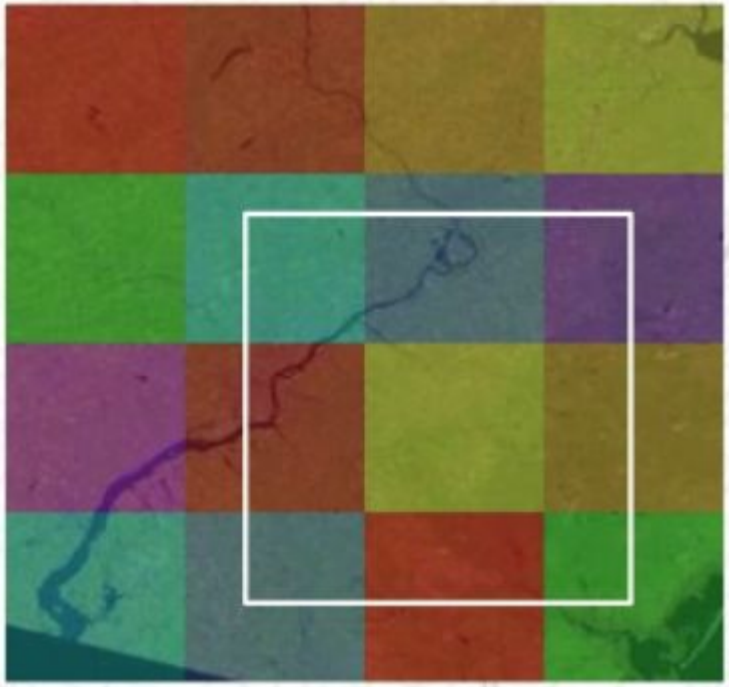


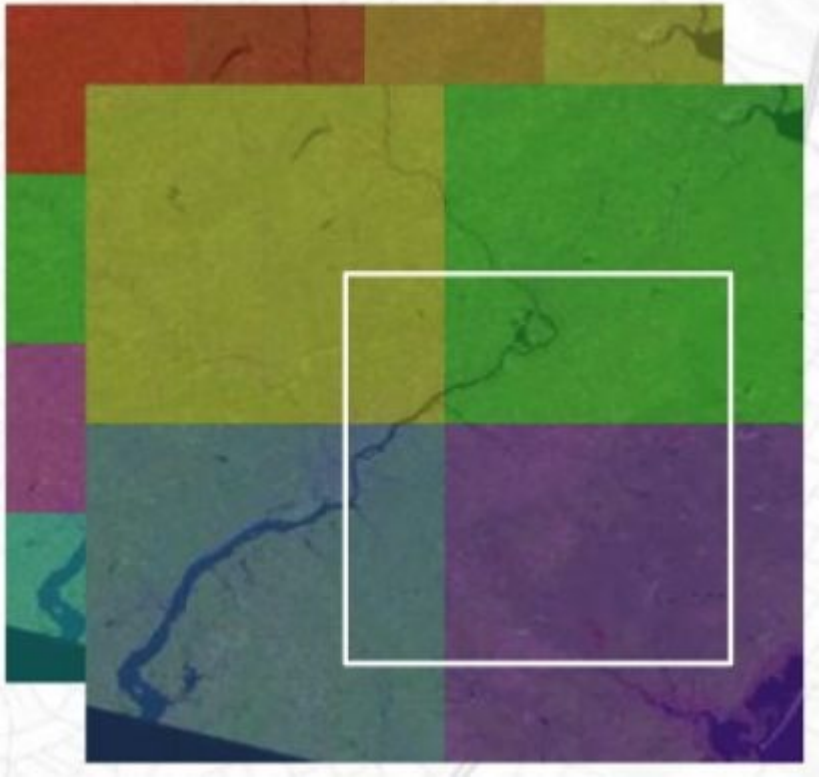
TIFF Tiled Segments = GOOD





TIFF Tiled Segments = GOOD





Using Earth Search for Sentinel-2

curl

$ curl -X POST https://earth-search.aws.element84.com/v0/search -d @payload.json

```
# payload.json
{
        "collections": ["sentinel-s2-l2a-cogs"],
        "datetime": "2020-01-01/now",
        "query": {
                "eo:cloud_cover": {
                        "lte": "10"
                }
        },
        "limit": 0
}
```

sat-search

$ python --version

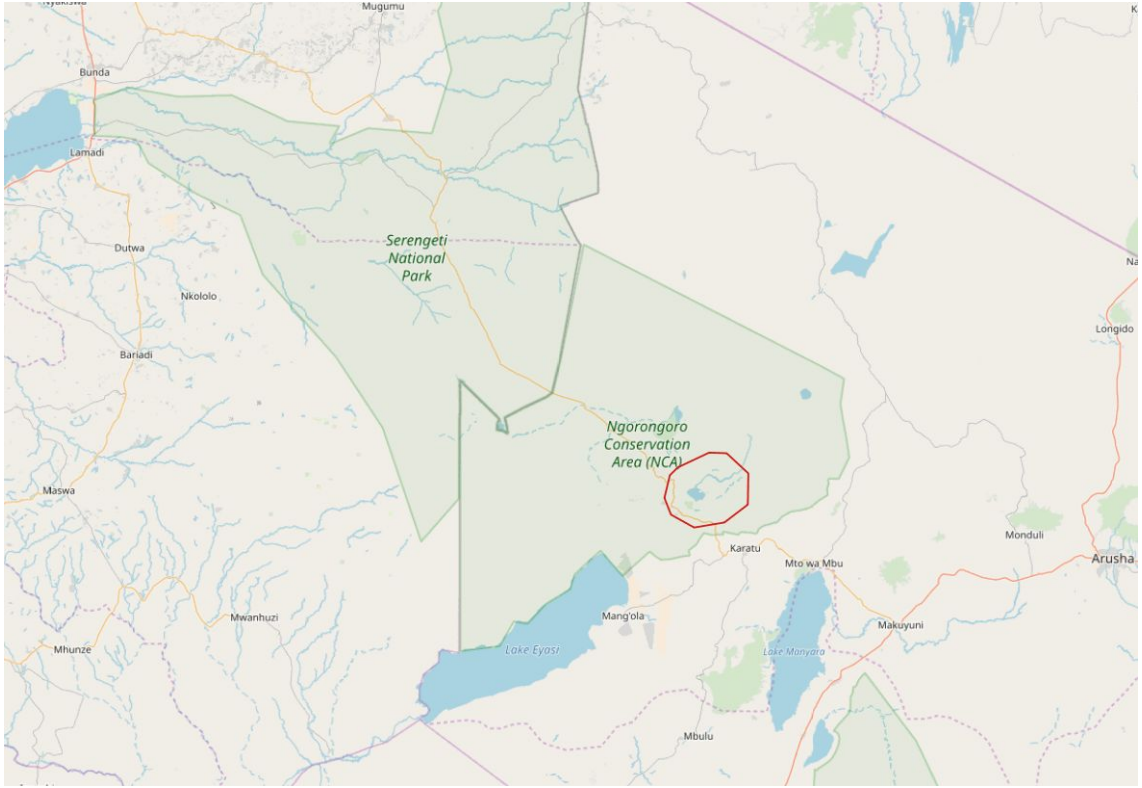
$ pip install sat-search==0.3.0-rc1

sat-fetch

$ docker run -v ${PWD}:/work -it satutils/sat-fetch:latest /bin/bash

```
$ sat-search search --intersects maine.geojson --found
1674 items found
$ clear
```

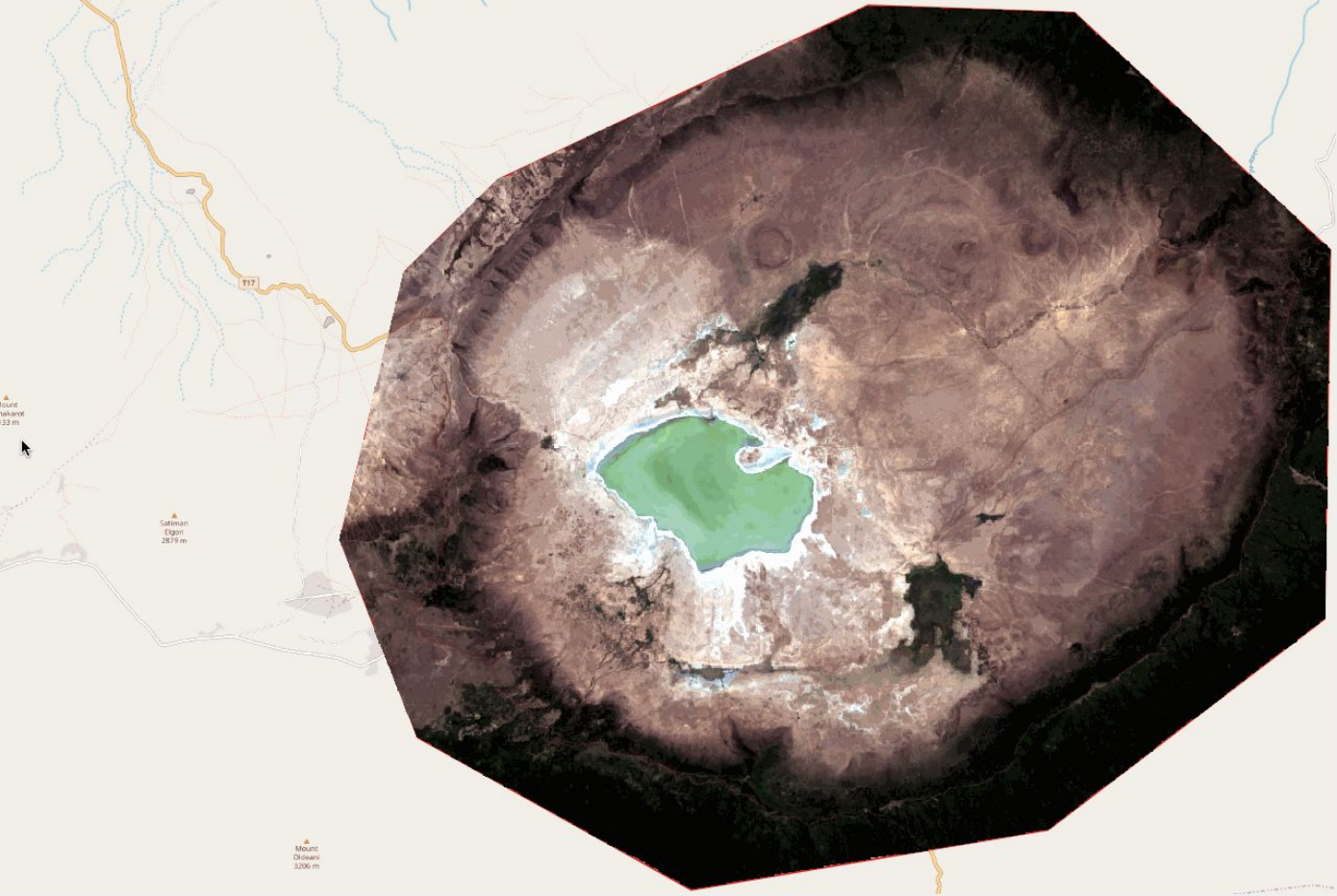
Ngorongoro crater: landsat-8 and sentinel-2



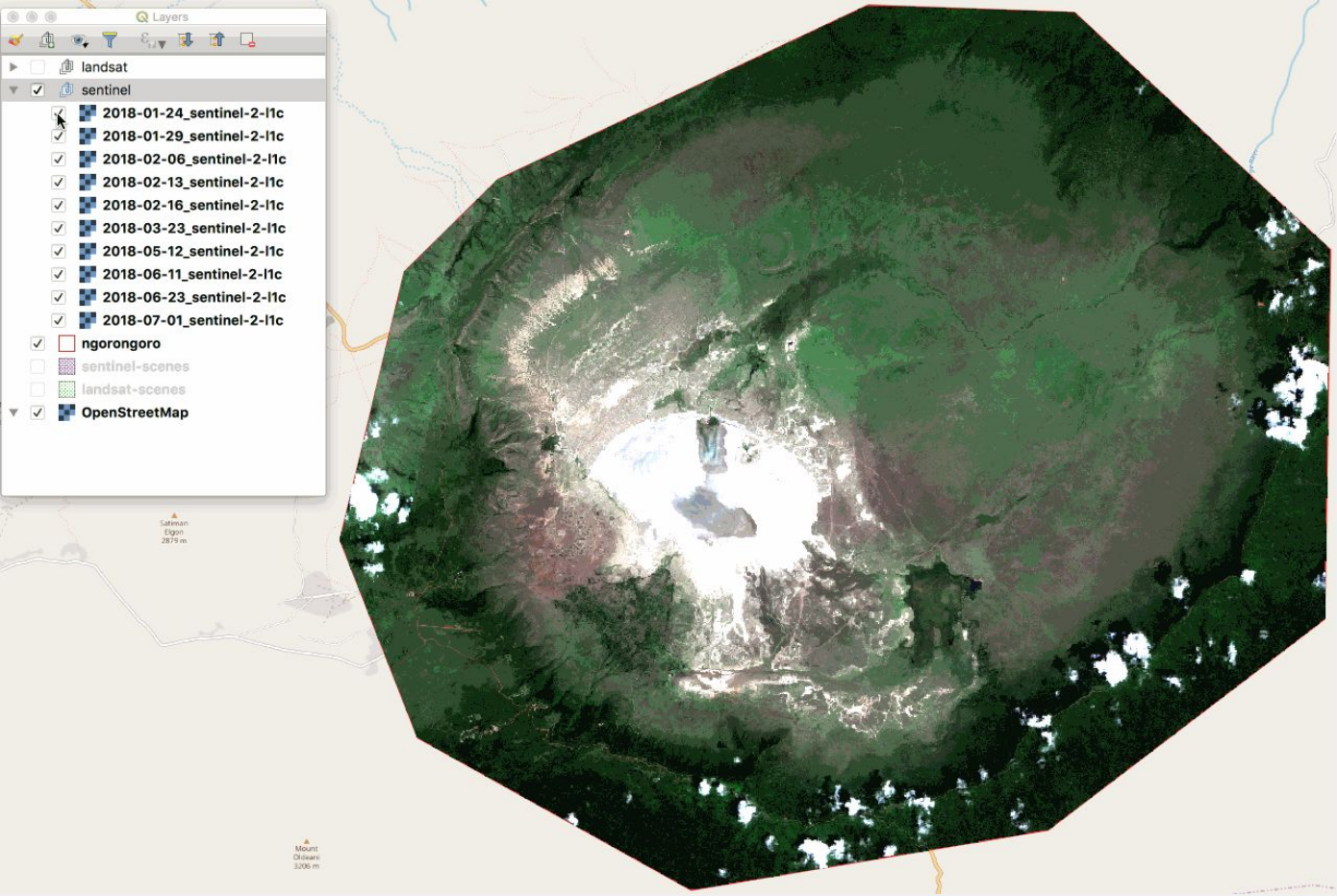
Ngorongoro crater: landsat-8 scenes

```
sat-search search
    \ --intersects ngorongoro.geojson --eo:cloud_cover 0/5 \
    -c landsat-8-l1 --save landsat-scenes.geojson \
    --print_md date eo:cloud_cover id
```

```
2014-12-11  5  LC08_L1TP_169062_20141211_20170416_01
2015-01-28  1  LC08_L1TP_169062_20150128_20170413_01
2015-02-13  0  LC08_L1TP_169062_20150213_20170413_01
2015-08-24  2  LC08_L1TP_169062_20150824_20170405_01
2015-09-09  1  LC08_L1TP_169062_20150909_20170404_01
2015-12-30  0  LC08_L1TP_169062_20151230_20170331_01
2016-02-16  0  LC08_L1TP_169062_20160216_20180526_01
2016-08-10  2  LC08_L1TP_169062_20160810_20170322_01
2016-10-13  0  LC08_L1TP_169062_20161013_20180526_01
2016-10-29  3  LC08_L1TP_169062_20161029_20170319_01
2017-01-17  0  LC08_L1TP_169062_20170117_20170311_01
2017-11-17  2  LC08_L1TP_169062_20171117_20171122_01
2018-02-05  0  LC08_L1TP_169062_20180205_20180221_01
2018-05-28  3  LC08_L1TP_169062_20180528_20180605_01
18 scenes found
```




```
$ time sat-fetch load landsat-scenes.geojson \
    --download red green blue
real 0m59.674s
```




```
$ time sat-fetch load sentinel-scenes.geojson \
    --download red green blue
```

For updates, releases, news:

@GeoSkeptic

Element 84

@element84